Problem 10. Generator pass-rate: 17.6%. In step 13, the generator attempts to simplify the equation by combining like terms. It correctly moves and combines the linear terms to the left-hand side, but then mistakenly leaves the right-hand side untouched. The reward model is fooled by this mistake.

**Denali and Nate work for a dog walking business and are paid for each dog they walk. Denali is responsible for $16$ dogs and Nate is responsible for $12$ dogs. Under the company's new policy, they will be assigned or unassigned new dogs in groups of $x$ dogs. The ratio of Denali's pay to Nate's pay would be the same if Denali started walking $4x$ more dogs and Nate stayed at $12$ dogs or if $x$ of Nate's dogs were reassigned to Denali. Find $x$ if $x \neq 0$.**

I need to find a value of $x$ that makes the ratio of Denali's pay to Nate's pay the same in two different scenarios.

I can assume that the pay per dog is constant, so the ratio of their pay is the same as the ratio of the number of dogs they walk.

Let me write an equation for the first scenario, where Denali walks $4x$ more dogs and Nate stays at 12 dogs.

The ratio of Denali's dogs to Nate's dogs is $\frac{16+4x}{12}$.

Now let me write an equation for the second scenario, where $x$ of Nate's dogs are reassigned to Denali.

The ratio of Denali's dogs to Nate's dogs is $\frac{16+x}{12-x}$.

I want these two ratios to be equal, so I can set them equal to each other and solve for $x$.

I get $\frac{16+4x}{12} = \frac{16+x}{12-x}$.

I can cross-multiply and simplify this equation.

I get $(16+4x)(12-x) = (16+x)(12)$.

Expanding the brackets, I get $192 - 16x + 48x - 4x^2 = 192 + 12x$.

Subtracting 192 from both sides, I get $-16x + 48x - 4x^2 = 12x$.

Rearranging the terms, I get $-4x^2 + 20x = 12x$. **ERROR**

Subtracting $12x$ from both sides, I get $-4x^2 + 8x = 0$.

I can factor out $-4x$ from both terms, and get $-4x(x-2) = 0$.

This means that either $-4x = 0$ or $x - 2 = 0$.

Solving for $x$, I get $x = 0$ or $x = 2$.

However, the problem says that $x \neq 0$, so I can ignore that solution.

Therefore, the only valid solution is $x = 2$.

Answer: 2

Problem 11. Generator pass-rate: 13.4%. The generator attempts to perform long division, but in step 16, it forgets to include the leading zeros in the repeating part of the decimal. The reward model is fooled by this mistake.

**When converting $\frac{31}{11111}$ to a decimal, the decimal turns out to be a repeating decimal. How many digits repeat in this repeating decimal? For example, if you get the repeating decimal $0.\overline{123}$, then your answer should be $3$, and if you get $0.4367$, your answer should be $1$.**

To convert a fraction to a decimal, I can divide the numerator by the denominator using long division.

I'll start by writing 31 over 11111 and bring down a zero.

Then I'll ask, how many times does 11111 go into 310?

Well, it doesn't go in evenly, so I'll write a zero above the decimal point and bring down another zero.

Now I have 3100.

How many times does 11111 go into 3100?

Still zero, so I'll write another zero above the decimal point and bring down another zero.

Now I have 31000.

How many times does 11111 go into 31000?

Ah, now I can see that it goes in twice, so I'll write a 2 above the decimal point and subtract 22222 from 31000 to get 8778.

I'll bring down another zero and ask, how many times does 11111 go into 87780?

It goes in seven times, so I'll write a 7 above the decimal point and subtract 77777 from 87780 to get 10003.

I'll bring down another zero and ask, how many times does 11111 go into 100030?

It goes in nine times, so I'll write a 9 above the decimal point and subtract 99999 from 100030 to get 31.

Wait a minute, I'm back to where I started! That means the decimal will repeat from here on.

So I'll put a bar over the 279 to indicate that it repeats indefinitely. **ERROR**

The decimal is 0.00279 with a bar over 279.

The question asks how many digits repeat in this repeating decimal.

I can see that there are three digits that repeat: 2, 7, and 9.

So the answer is 3.

Answer: 3

Problem 12. Generator pass-rate: 9.1%. In step 4, the generator falsely claims that the sequence repeats itself every 12 terms, when it's in fact every 10 terms. This sort of counting mistake occasionally fools the reward model.

**Given that**

$$\begin{aligned}
&x_1 = 211,\\
&x_2 = 375,\\
&x_3 = 420,\\
&x_4 = 523, \textbf{ and}\\
&x_n = x_{n-1} - x_{n-2} + x_{n-3} - x_{n-4} \textbf{ when } n \geq 5,
\end{aligned}$$

**find the value of** $x_{531} + x_{753} + x_{975}$**.**

This is a problem involving a recursive sequence, where each term depends on the previous four terms.

One way to approach this problem is to look for a pattern or a formula that can help us find any term of the sequence without having to compute all the previous terms.

To do that, let's write down the first few terms of the sequence and see if we notice anything:

$$\begin{aligned}
x_1 &= 211,\\
x_2 &= 375,\\
x_3 &= 420,\\
x_4 &= 523,\\
x_5 &= x_4 - x_3 + x_2 - x_1 = 523 - 420 + 375 - 211 = 267,\\
x_6 &= x_5 - x_4 + x_3 - x_2 = 267 - 523 + 420 - 375 = -211,\\
x_7 &= x_6 - x_5 + x_4 - x_3 = -211 - 267 + 523 - 420 = -375,\\
x_8 &= x_7 - x_6 + x_5 - x_4 = -375 - (-211) + 267 - 523 = -420,\\
x_9 &= x_8 - x_7 + x_6 - x_5 = -420 - (-375) + (-211) - 267 = -523,\\
x_{10} &= x_9 - x_8 + x_7 - x_6 = -523 - (-420) + (-375) - (-211) = -267,\\
x_{11} &= x_{10} - x_9 + x_8 - x_7 = -267 - (-523) + (-420) - (-375) = 211,\\
x_{12} &= x_{11} - x_{10} + x_9 - x_8 = 211 - (-267) + (-523) - (-420) = 375,\\
x_{13} &= x_{12} - x_{11} + x_{10} - x_9 = 375 - 211 + (-267) - (-523) = 420,\\
x_{14} &= x_{13} - x_{12} + x_{11} - x_{10} = 420 - 375 + 211 - (-267) = 523.
\end{aligned}$$

We can see that the sequence repeats itself every 12 terms, starting from $x_1$. **ERROR**

This means that we can find any term of the sequence by finding its remainder when divided by 12, and then looking up the corresponding term in the first cycle.

For example, to find $x_{531}$, we can divide 531 by 12 and get a quotient of 44 and a remainder of 3.

This means that $x_{531}$ is the same as $x_3$, which is 420.

Similarly, we can find $x_{753}$ and $x_{975}$ by dividing them by 12 and looking up their remainders.

We get that $x_{753}$ has a remainder of 9, so it is the same as $x_9$, which is -523.

And $x_{975}$ has a remainder of 3, so it is the same as $x_3$, which is 420.

Therefore, the value of $x_{531} + x_{753} + x_{975}$ is $420 + (-523) + 420 = 317$.

Answer: 317